Syllabus: Psychrometric properties and processes, sensible and latent heat loads, need for ventilation, infiltrated air, heat load concepts. Air cooler (evaporative cooling), window, split, summer, winter, year-round and central air conditioning systems.

I. Introduction to Air Conditioning

- **Air conditioning** = simultaneous control of temperature, humidity, air motion (velocity) and cleanliness (purity) of air in an enclosed space, to meet the requirement of occupants or of a process.
- **Comfort AC:** homes, offices, theatres, vehicles (≈ 22-26 °C, 40-60 % RH). **Industrial AC:** process control – textile mills, pharma, printing, computer rooms, operation theatres.
- Four basic functions: **cooling / heating, humidifying / dehumidifying, air cleaning (filtering), air distribution (and ventilation).**

Psychrometry – properties of moist air

Moist air = dry air + water vapour (Dalton's law: $p = p_a + p_v$). Properties used:

- **Dry bulb temperature (DBT)** – temperature by ordinary thermometer.
- **Wet bulb temperature (WBT)** – temperature of a wetted bulb in moving air; measures evaporative cooling potential. **WBT ≤ DBT.**
- **Dew point temperature (DPT)** – temperature at which vapour begins to condense when air is cooled at constant pressure; $p_{sat}(DPT) = p_v$.
- **Relative humidity ()** = p_v/p_{sat} at the DBT.
- **Specific (absolute) humidity (W)** = mass of vapour / mass of dry air.

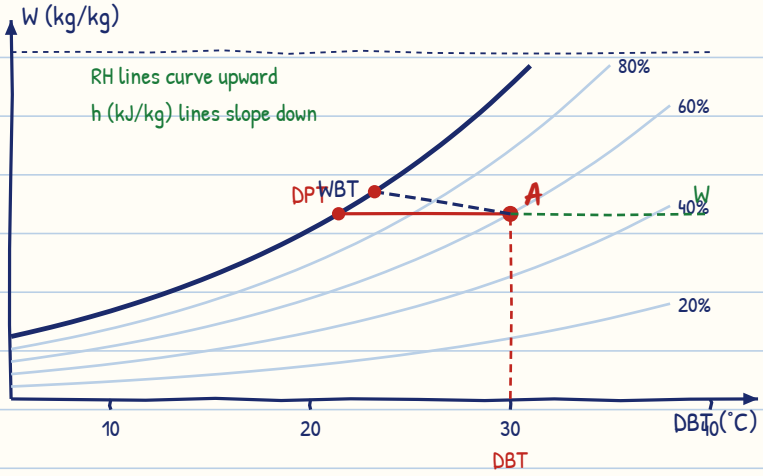
$$W = 0.622\ p_v / (p - p_v)$$

- **Degree of saturation** = W / W_s (at same DBT).
- **Specific enthalpy** of moist air per kg dry air:

$$h = 1.005\ t + W\ (2500 + 1.88\ t)\ kJ/kg\ d.a.$$

- **Specific volume** $v = R_a\ T\ (1 + 1.608\ W) / p \approx 0.83 - 0.95\ m^3/kg\ d.a.$

Note: For saturated air DBT = WBT = DPT. For unsaturated air DBT > WBT > DPT.



Point A: 30 °C DBT, 60 % RH
- horizontal line → DPT (21.4 °C)
- constant W line = dew point line
- sloping line up-left → WBT (≈ 23.2)
- enthalpy ≈ 71 kJ/kg
- v ≈ 0.88 m³/kg

Example 1 – properties by calculation

Air at 30 °C, 60 % RH, $p = 101.325$ kPa. $p_{sat}(30\ °C) = 4.246$ kPa.

$$p_v = 0.6 \times 4.246 = 2.548 \text{ kPa} \;;\; \text{DPT: } p_{sat} = 2.548 \rightarrow \mathbf{21.4\ °C}$$

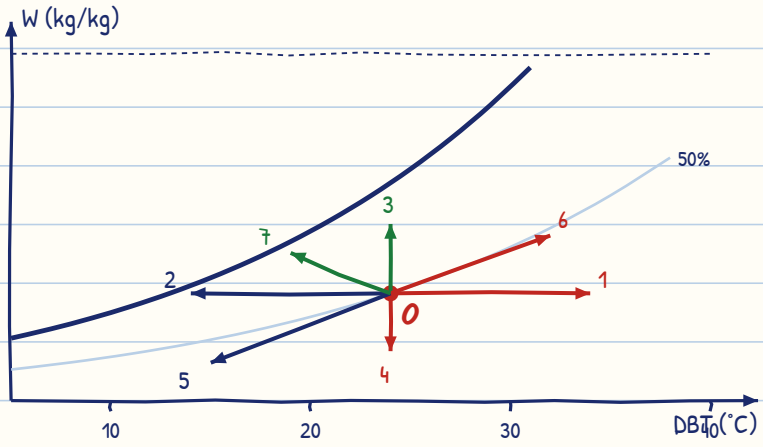
$$W = 0.622 \times 2.548 \,/\, (101.325 - 2.548) = \mathbf{0.01604\ kg/kg}$$

$$h = 1.005 \times 30 + 0.01604\,(2500 + 1.88 \times 30) = 30.15 + 41.0 = \mathbf{71.2\ kJ/kg}$$

$$v = 0.287 \times 303\,(1 + 1.608 \times 0.01604)/101.325 = \mathbf{0.88\ m^3/kg}$$

2. Psychrometric Chart and Processes

- The **psychrometric chart** plots W (vertical, right) against DBT (horizontal). Lines on it: constant DBT (vertical), constant W (horizontal), constant RH (curves), constant WBT (sloping down to right), constant enthalpy (nearly same as WBT), constant specific volume (steeper slope).
- Any two independent properties fix the state.



1 sensible heating

2 sensible cooling

3 humidification

4 dehumidification

5 cooling + dehumidification

6 heating + humidification

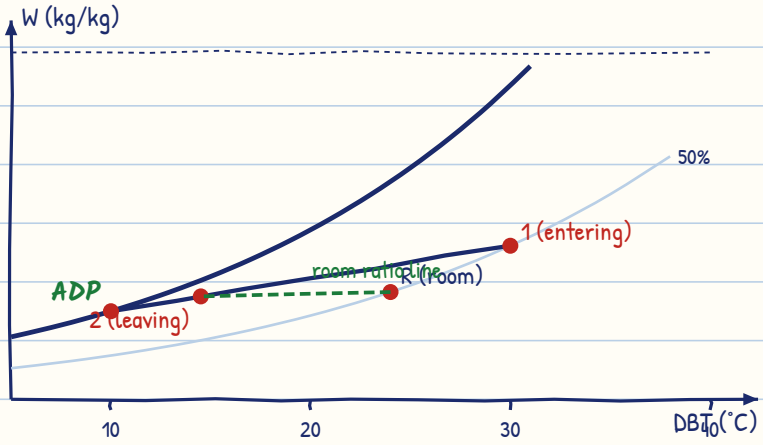
7 evaporative cooling (WBT)

+ mixing: point on line joining

Process equations (m_a = dry air flow, kg/s)

- **Sensible heating / cooling** (W = const): $Q = m_a\, c_{pm}\, \Delta t = m_a\,(1.005 + 1.88\,W)\,\Delta t \approx 1.022\, m_a\, \Delta t$ kW. With volume flow V (m^3/s): **$Q_s = 1.23\, V\, \Delta t$** kW.
- **Humidification** (heating and adding steam): latent heat $Q_L = m_a\,(W_2 - W_1)\,h_{fg}$; **$Q_L = 3012\, V\, \Delta W$** kW ($h_{fg} \approx 2500$, $\rho = 1.2$).
- **Cooling and dehumidification:** air cooled below its dew point; $Q = m_a\,(h_1 - h_2) - m_w\, h_f$; condensate $m_w = m_a\,(W_1 - W_2)$.
- **Heating and humidification (winter AC):** $Q_{heat} = m_a\,(h_2 - h_1)$.
- **Adiabatic mixing of two streams:** $h_3 = (m_1h_1 + m_2h_2)/(m_1 + m_2)$, $W_3 = (m_1W_1 + m_2W_2)/(m_1 + m_2)$ – the mixed state lies on the straight line joining 1 and 2.
- **Adiabatic saturation / evaporative cooling:** h ≈ constant along WBT line; DBT falls, W rises.
- **Chemical dehumidification:** air passes over silica gel / calcium chloride – W falls, DBT rises (h ≈ const).

Cooling coil – ADP and bypass factor



ADP = coil surface temp.

Bypass factor

$BF = (t_2 - t_{ADP}) / (t_1 - t_{ADP})$

Contact factor = 1 – BF

SHF = SH / (SH + LH)

- **Apparatus Dew Point (ADP):** the average coil surface temperature. Air leaving a coil is a mix of air that touched the coil (at ADP) and air that bypassed.

$$BF = (t_2 - t_{ADP}) / (t_1 - t_{ADP}) \text{ ; Contact factor } CF = 1 - BF$$

- BF is smaller for deeper coils, more fins, lower air velocity. Typical BF: 0.1-0.3.
- **Sensible Heat Factor:** $SHF = Q_s / (Q_s + Q_L)$. **RSHF** (room), **GSHF** (grand = room + ventilation load), **ESHF** (effective, at coil).
- The **room ratio line** (slope fixed by RSHF) is drawn from the room state; its intersection with the saturation line gives the required ADP.

How to plot a state on the chart (steps)

- Mark the DBT on the horizontal axis and go vertically up; read the intersection with the RH curve (or WBT line) – that is the state point.
- From the point go horizontally to the right to read W; go along the sloping line to the left to read WBT / enthalpy scale; horizontally left to the saturation curve for DPT.
- For a process draw a straight line between the two states; use **protractor (SHF scale)** on the chart for the room ratio line.
- Condition lines: comfort zone ≈ 22-26 °C DBT and 40-60 % RH (the "bounded zone").

3. Sensible and Latent Heat Loads

Sensible heat loads

- **Solar radiation** through windows (largest): $Q = A \times SHGF \times SC$.
- **Transmission** through walls, roof, floor, glass: $Q = U\ A\ (CLTD)$ or $U\ A\ (t_o - t_i)$.

- **Internal:** lights (W × 1.25 for fluorescent ballast), people (≈ 75 W sensible per person seated), motors and equipment, cooking appliances.
- **Infiltration and ventilation** air (sensible portion): $Q_s = 1.23\ V\ (t_o - t_i)$.

Latent heat loads

- **People** (≈ 55 W latent per person), **infiltration / ventilation** air moisture: $Q_L = 3012\ V\ (W_o - W_i)$, cooking, steam from processes, evaporation from wet surfaces, door openings.

Source	Sensible	Latent
Occupants (seated)	75 W/person	55 W/person
Fluorescent lights	1.25 × W	–
Equipment (computers)	rated W × use factor	–
Walls, roof, glass	U A ΔT, solar	–
Infiltration / ventilation	1.23 V Δt	3012 V ΔW
Cooking / steam	some	large

Solar and transmission load formulae

Solar gain through glass: $Q = A \times SHGF_{max} \times SC \times CLF$ (A = glass area, SC = shading coefficient)

Wall / roof: $Q = U\ A\ (CLTD)$; $U = 1 / (1/h_o + \Sigma\ L/k + 1/h_i)$ W/m^2K

- Typical U: brick wall 2.0, insulated roof 0.4–0.6, single glass 5.8, double glass 2.8. CLTD (cooling load temperature difference) accounts for time lag due to thermal storage.
- **Diversity factor:** not all loads peak together; people, lights → multiply by use factor 0.7–0.9.

Total load

Room sensible heat (RSH) + room latent heat (RLH) = Room total heat (RTH); RSHF = RSH / RTH

Grand total heat = RTH + outdoor air load (OASH + OALH) + supply duct gain, fan heat, leakage.

- Add 5–10 % safety factor. Cooling TR = Grand total heat (kW)/3.5167.

4. Need for Ventilation and Infiltrated Air

Ventilation

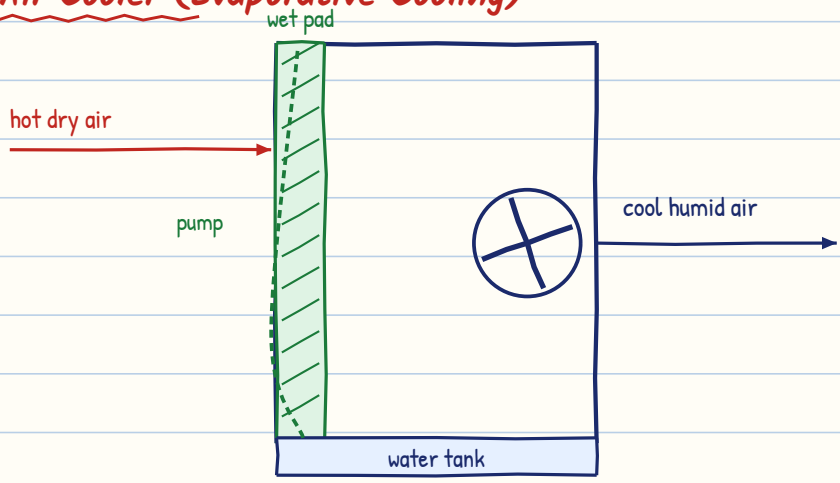
- **Needed:** to supply oxygen, dilute CO_2 (limit ≈ 1000 ppm) and body odours, smoke, VOCs; to maintain positive pressure in clean rooms.
- Fresh air is mixed with the return air; typical requirement ≈ 7-10 L/s (≈ 0.4-0.6 m^3/min) per person for offices; ASHRAE 62.1 ≈ 2.5 L/s per person + 0.3 L/s per m^2 floor.
- **Fresh air = 10-30 %** of supply air. Too much fresh air adds load; too little affects health ("sick building").
- Methods: natural (windows, stack), mechanical (supply fan, exhaust fan), heat recovery wheel (ERV) to save energy.

Infiltration

- Uncontrolled leakage of outside air through cracks around doors and windows, door openings, and the **stack effect** (hot air rises) and **wind pressure**.
- **Air change method:** V = (ACH × room volume)/3600 m^3/s. Typical 0.5-1 ACH for houses, 1-2 for shops with door use.
- **Crack method:** V = (crack length) × (leakage per m) at the design pressure difference.
- Reduce by weather stripping, double doors, air curtain, and keeping the room under slight positive pressure.

Infiltration load: $Q_s = 1.23\ V\ (t_o - t_i)$, $Q_L = 3012\ V\ (W_o - W_i)$

5. Air Cooler (Evaporative Cooling)



Desert cooler (direct)

Process: sensible → latent

$t_2 = t_1 - \eta\ (t_1 - t_{wb})$

η = 70-90 %

DBT ↓ , W ↑ , h ≈ const

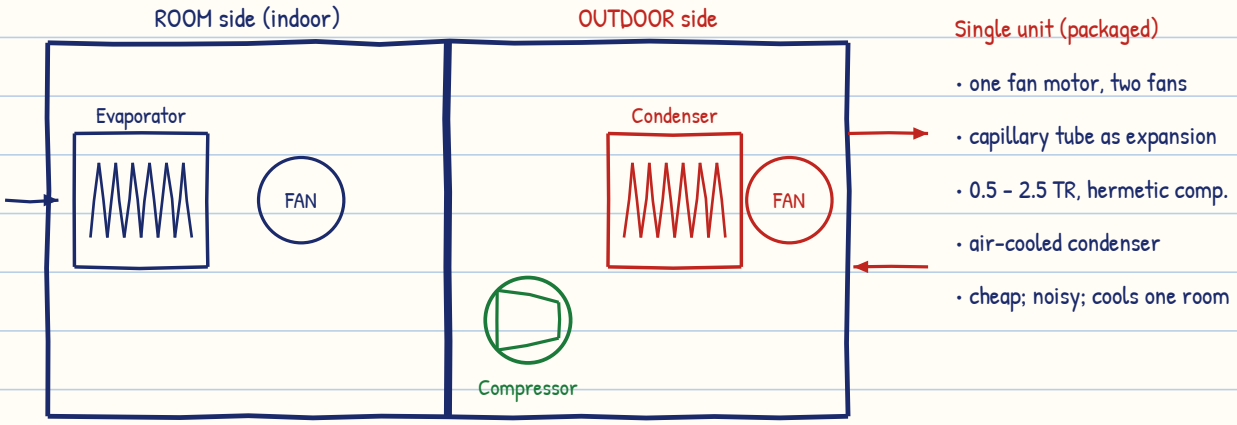
- Water is circulated by a small pump over a porous pad (khus / honeycomb cellulose). Warm dry air passing over the wet pad evaporates water, so the **air loses sensible heat** to supply latent heat – DBT falls, humidity rises (h ≈ const).

> Saturation effectiveness $\eta = (t_1 - t_2) / (t_1 - t_{wb})$

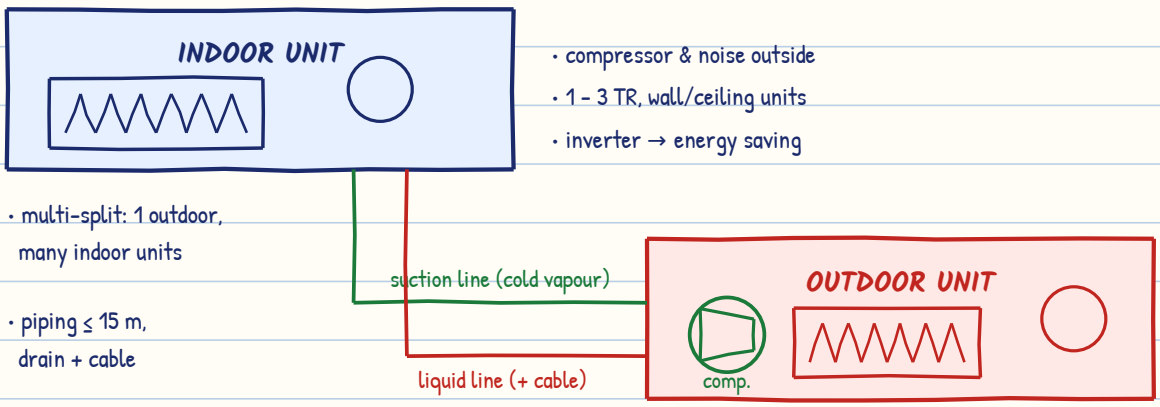
- Example: outside 40 °C, WBT 24 °C, $\eta = 0.8 \rightarrow t_2 = 40 - 0.8 \times 16 =$ **27.2 °C**.
- **Suitable for hot and dry climates** (Rajasthan, Deccan); poor in humid coastal regions. Low cost (power ≈ 1/10 of AC), needs ventilation (open window) to exhaust humid air. **Indirect** evaporative cooler uses a heat exchanger – no humidity addition. **Two-stage** (indirect + direct) can cool below WBT.

6. Types of Air Conditioning Systems

Window air conditioner

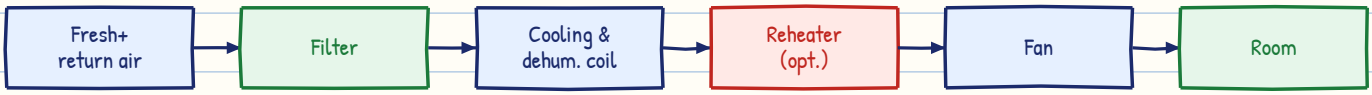


Split air conditioner

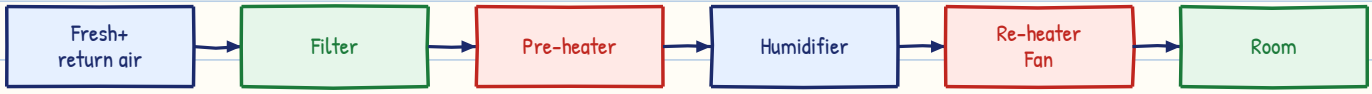


Summer, Winter and Year-round systems

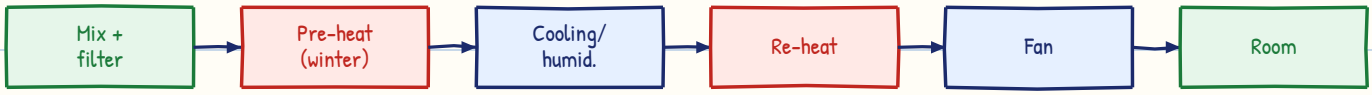
Summer AC



Winter AC

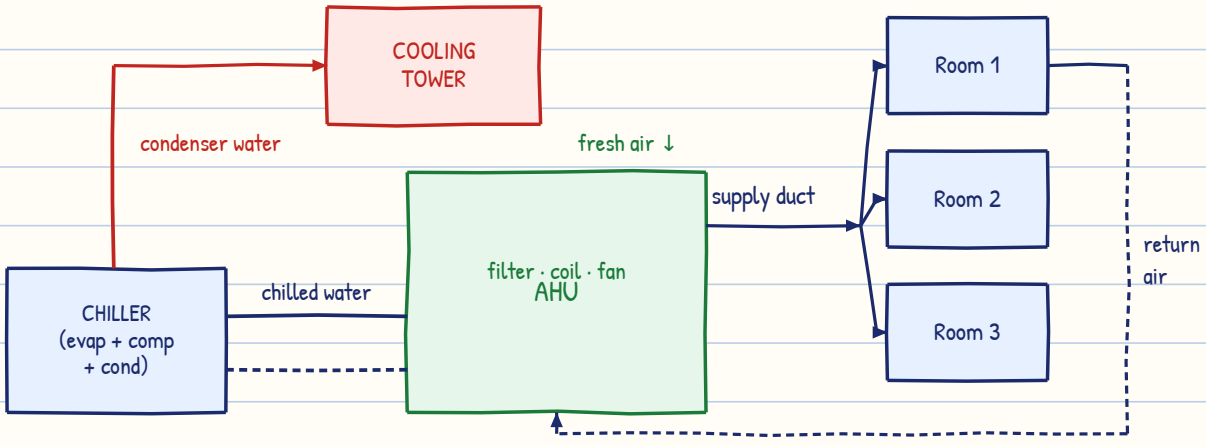


Year-round



- **Summer AC:** cooling and dehumidification with the cooling coil, then reheating if needed. Typical room 24-26 °C and 50 % RH. Load: RSHF line and ADP.
- **Winter AC:** heating and humidification (preheat coil → humidifier / steam sprayer → reheat coil). Room 21-22 °C, 40 % RH. Pre-heating avoids freezing of water spray.
- **Year-round AC:** both cooling/dehumidifying and heating/humidifying coils in one plant, with controls for the season, fresh-air mixing and reheat. Needed in places with both hot and cold seasons.

Central air conditioning



- Large buildings, hotels, hospitals, malls: **chiller plant** (VCR or VAR) makes chilled water (7 → 12 °C) circulated to **Air Handling Units (AHU)** or fan coil units; AHU conditions air and ducts distribute it. Heat rejection through a **cooling tower**.

- **Types:** all-air (constant air volume, VAV), air-water (fan coil + primary air), all-water, **VRF/VRV** (variable refrigerant flow).
- **Pros:** centralised maintenance, better humidity control, one plant for many zones, quiet in rooms. **Cons:** high initial cost, large duct space, skilled operators needed.

Type	Capacity	Pros	Cons
Window	0.5 – 2.5 TR	Cheap, simple	Noisy, one room
Split	1 – 5 TR	Quiet, flexible	Pipe length limit
Packaged	3 – 25 TR	Ducted, easy install	Space outside
Central	> 25 TR	Large buildings, control	Costly, ducts

6A. Air Distribution, Ducts and AHU

- **Supply outlets:** grille, register (with damper), ceiling diffuser (square / round / linear slot). Aim: uniform temperature, velocity < 0.25 m/s in occupied zone (draught-free). **Throw** = distance air travels from outlet to terminal velocity (0.25 m/s); **drop**, **spread**.
- **Return air** through grilles at low level/ceiling. Supply-return balanced so that room pressure is slightly positive.
- **Duct design:** equal friction method (≈ 0.8-1.0 Pa/m), velocity reduction, static regain; main duct velocity 6-9 m/s, branches 3-5 m/s; rectangular / round GI sheet with insulation and vibration-free flexible connections.

AHU components

- Mixing box with dampers (fresh + return) → pre-filter → cooling coil (chilled water / DX) → heating coil → humidifier → supply fan (centrifugal) → filters (fine) → supply duct. Drain tray for condensate.
- **Fan coil unit (FCU):** small coil + fan in the room, chilled / hot water supply; **VAV:** supply volume varies with load (saves fan energy); **VRF:** refrigerant flow varied by inverter compressor to many indoor units.

Direct expansion (DX)	Chilled water
Refrigerant circulated to coil	Water circulated, primary VCR/VAR at plant
Up to ≈ 25 TR, packaged	Large buildings, > 25 TR
Short pipe, cheap	Long distance, better control

Controls

- Thermostat (DBT), humidistat (RH), pressure cut-outs, damper actuators; BMS for large buildings. Set-points: 24 ± 1 °C and 50 ± 5 %.

Step-wise design procedure for summer air conditioning

- **Step 1:** fix outdoor design (e.g. 35 °C DBT, 24 °C WBT) and indoor design (24 °C, 50 %) conditions.
- **Step 2:** calculate room sensible (RSH) and latent (RLH) loads → RSHF.
- **Step 3:** draw the room ratio line through the room state R; choose supply temperature difference (8-12 K) → supply state S; or find ADP where the line cuts the saturation curve.
- **Step 4:** supply air $V = RSH / (1.23 (t_R - t_S))$. Fresh air fraction chosen from the ventilation rule; mix with return air to get state M.
- **Step 5:** coil enters at M, leaves at S (along the line to ADP): find BF; coil load = $m_a (h_M - h_S)$ → TR.
- **Step 6:** select fan, ducts, filters, chiller, cooling tower; add reheater if S is too cold.

Space	Fresh air (L/s per person)	Design condition
Office	7 - 10	24 °C, 50 %
Theatre	5 - 8	24 °C, 55 %
Restaurant	10 - 15	24 °C, 55 %
Hospital ward	12 - 15	23 °C, 50 %
Operation theatre	100 % fresh, 20 ACH	21 °C, 50 %

Energy-saving measures

- Use inverter compressors, VAV and VRF; energy recovery ventilators; economiser (free cooling using cool outside air); insulation, double glazing, shading; set point 24-26 °C; clean filters and coils.

7. Numerical Problems

Problem 1 (sensible heating)

2 m^3/s of air is heated from 10 °C to 30 °C. $Q = 1.23 \times 2 \times 20 =$ **49.2 kW**.

Problem 2 (mixing)

2 kg/s air at 25 °C, $W = 0.010$ is mixed with 1 kg/s at 35 °C, $W = 0.020$.

$t_3 = (2\times25 + 1\times35)/3 = \mathbf{28.3\ °C}$; $W_3 = (2\times0.010 + 1\times0.020)/3 = \mathbf{0.0133\ kg/kg}$

Problem 3 (supply air quantity)

Room 24 °C, 50 % RH, RSH = 20 kW and RLH = 5 kW → RSHF = 20/25 = **0.8**. Supply air at 14 °C:

$$V = RSH / (1.23\ \Delta t) = 20 / (1.23 \times 10) = \mathbf{1.63\ m^3/s} \approx 5850\ m^3/h$$

Problem 4 (bypass factor)

Coil entering air 28 °C, leaving 14 °C, ADP 10 °C: BF = (14 − 10)/(28 − 10) = **0.222**, CF = 0.778.

Problem 5 (heat load of a 100 m² office, 20 persons)

People: sensible 20×75 = 1.5 kW ; latent 20×55 = 1.1 kW

Lights 1.0 kW × 1.25 = 1.25 kW ; Equipment 2 kW ; Envelope 5 kW

Fresh air: 20 × 7.5 L/s = 0.15 m³/s ; 35 °C → 24 °C, ΔW = 0.0075

OA sensible = 1.23×0.15×11 = 2.03 kW ; OA latent = 3012×0.15×0.0075 = 3.39 kW

Total sensible = 1.5+1.25+2+5+2.03 = 11.78 kW ; Total latent = 1.1+3.39 = 4.49 kW

Grand total = 16.27 kW = **4.6 TR** ; SHF = 11.78/16.27 = 0.72

Problem 6 (evaporative cooler)

t_1 = 38 °C, t_{wb} = 22 °C, η = 85 % → $t_2 = 38 - 0.85 \times 16 = \mathbf{24.4\ °C}$.

Problem 7 (cooling coil load)

1.5 kg/s of air enters a coil at h_1 = 60 kJ/kg, W_1 = 0.0140 and leaves at h_2 = 40 kJ/kg, W_2 = 0.0090.

Condensate = 1.5 (0.0140 − 0.0090) = 0.0075 kg/s = **27 kg/h**

$Q = 1.5\ (60 - 40) - 0.0075 \times 58.8 = 30 - 0.44 =$ **29.6 kW = 8.4 TR**

Problem 8 (ventilation load of a theatre)

200 persons at 7.5 L/s = 1.5 m^3/s outdoor air, 35 °C to 24 °C, $\Delta W = 0.0075$:

OA sensible = 1.23×1.5×11 = 20.3 kW ; OA latent = 3012×1.5×0.0075 = 33.9 kW ; total = **54.2 kW = 15.4 TR**

Note: Ventilation load is often 25–40 % of the total – energy recovery wheel is used to reduce it.

Problem 9 (infiltration)

Room 5 × 4 × 3 = 60 m^3 with 1.5 ACH → V = 60×1.5/3600 = 0.025 m^3/s.

$Q_s = 1.23 \times 0.025 \times 11 = 0.34$ kW ; $Q_L = 3012 \times 0.025 \times 0.0075 = 0.56$ kW ; total = **0.9 kW**

Short answers (2 marks)

- **What is dew point?** The temperature at which water vapour starts condensing when air is cooled at constant pressure and constant W.
- **Why is SHF important?** It fixes the slope of the room ratio line and the ADP.
- **What is infiltration?** Uncontrolled leakage of outside air into the conditioned space.
- **Evaporative cooling – process line?** Along constant WBT / constant enthalpy.
- **Which AC for large hotels?** Central AC with chiller, AHU and cooling tower.

8. Quick Revision and Exam Questions

- $W = 0.622\ p_v/(p - p_v)$; $h = 1.005t + W(2500 + 1.88t)$; $Q_s = 1.23\ V\ \Delta t$; $Q_L = 3012\ V\ \Delta W$.
- SHF = SH/(SH+LH); BF = $(t_2 - t_{ADP})/(t_1 - t_{ADP})$; evaporative $\eta = (t_1 - t_2)/(t_1 - t_{wb})$.
- Window = packaged; split = indoor + outdoor; central = chiller + AHU + cooling tower.

Likely questions

- Define DBT, WBT, DPT, RH, specific humidity; show on psychrometric chart. (short)
- Explain psychrometric processes with chart and equations. (long)
- Define SHF, RSHF, GSHF, ADP and bypass factor. (short / long)
- Components of cooling load; need for ventilation and infiltration. (long)

- Explain with schematic: summer, winter, year-round AC. (long)
- Window, split and central AC – compare. (long)
- Evaporative cooling – principle, effectiveness, application. (short)